Estimation or Experimentation?

Demand Learning Under Partial Price Transparency

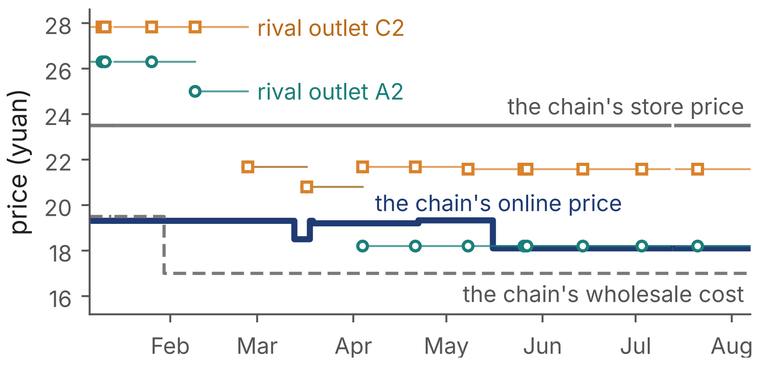
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Full paper
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1 One product in one trade area

A pharmacy chain sells a 15 g tube of desonide cream 0.05% in its store and on a delivery platform. On the platform, its staff see the online prices of nearby rival outlets, but not their costs or store prices, and they must decide which rival listing is the same product as their own.



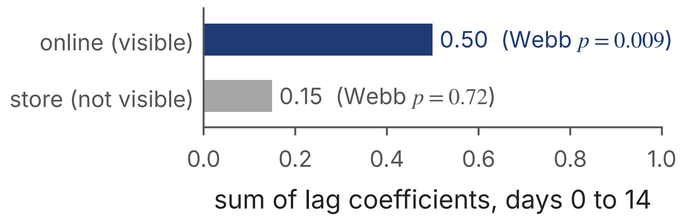
The chain's listed prices and wholesale cost and the recorded online prices of two rival outlets, 6 January to 6 August 2026. After the cost fell, both rivals' recorded prices fell, and the chain cut its online price to just below the cheaper one. Its store price never changed.

The rival price that a seller records is its **quote**. The gap between the quote and the rival's actual price is the **wedge**, and the inverse variance of the wedge is the quote's **precision**. A platform that standardizes product descriptions, or a seller that matches listings more carefully, raises precision.

Question. A regression of sales on the own price and a noisy quote understates price sensitivity. A price experiment does not. Do competing sellers that choose by profit replace the regression, and what does their choice imply for prices?

2 Evidence from the pharmacy panel

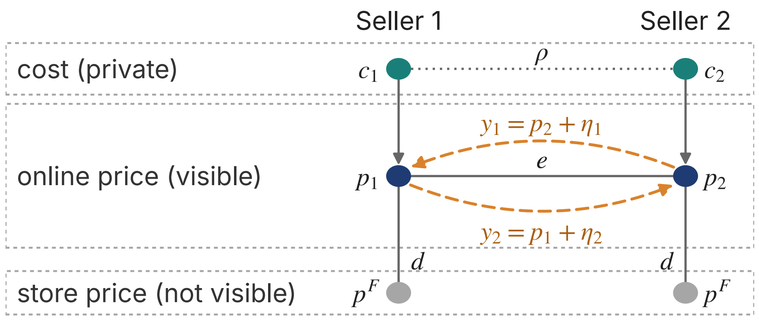
One city · 6 trade areas · 3,937 products · 6 chain and 14 rival outlets · 6 January to 6 August 2026



Regressions of rival online log price changes on the chain's daily log price changes in each channel, with product and date fixed effects. Wild cluster bootstrap by trade area.

- Rival online prices move with the chain's **visible** online price changes, and not significantly with its store price changes, which rivals do not see.
- Store prices are sticky: 82% of the chain's product and store pairs never changed the store price, against 60% for the online price.
- The chain's median new online price is 0.3% below the cheapest rival record in the trade area.
- Online sales respond to the own price: -0.41 ± 0.27 log units per unit log price change over two weeks (Webb $p = 0.054$).
- Both sides reprice by judgment. The panel describes the setting and does not test the choice.

3 The model



- Demand $q_i = \alpha - b\,p_i + d\,p^F + e\,p_j + \epsilon_i$, with $\omega = e/(2b)$.
- Quote $y_i = p_j + \eta_i$, $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$. Precision is $1/\sigma_\eta^2$.
- Rule $p_i = k_i + g_i\,\theta_i + w_i\,y_i$, with quote weight w_i.

A **learning procedure** maps the seller's data to its rule:

- **Estimation (M)**
  Regress sales on the own price and the quote. Price optimally under the fitted demand.
- **Experimentation (X)**
  Perturb the price at random and measure the response of sales. Price on the measured response.
- **Benchmark (R)**
  Know the own-price slope b. Price with the rival's price held fixed.

A **stopping point** is a pair of rules that the procedures return from the data that the rules generate. Sellers choose procedures by **long-run profit**, the profit at the stopping point.

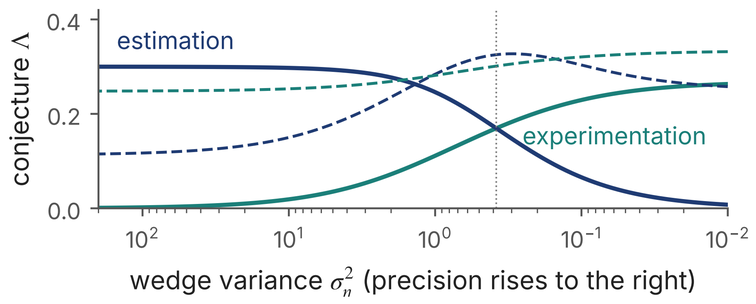
4 A procedure determines a conjecture

A seller that acts on the own-price slope β_i prices as if a unit increase in its price raised the rival's price by its **conjecture** $\Lambda_i = (b - \beta_i)/e$.

Lemma 1. At every stopping point, $\Lambda_i = 0$ under R, $\Lambda_i = w_j$ under X, and under M

$$\Lambda_i^M = t_i \left(\Lambda_i^X + \rho\, \frac{g_j}{g_i} \right),$$

where t_i is the share of the wedge in the variance of the quote. Each mean price rises with either seller's conjecture and is at or above the Nash level.



Solid: the conjectures when both sellers use the same procedure, at the benchmark. Dashed: the conjecture with the highest long-run profit against a rival that estimates (navy) or experiments (teal). Dotted: the switch precision.

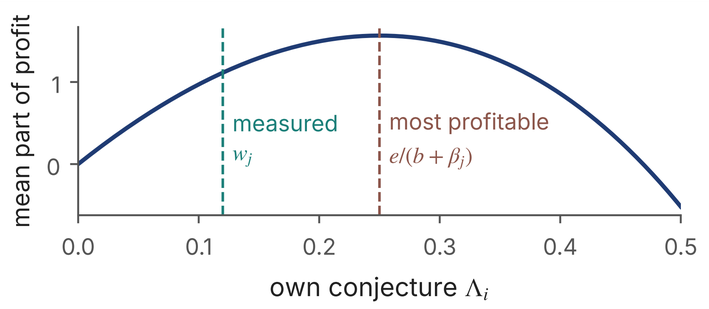
- Precise quotes make the conjecture of X large, because sellers put much weight on them, and that of M small, because the control works. Imprecise quotes do the reverse.

5 Best today, not in the long run

Theorem 1. Against the rival's **current** rule, X earns more than every other affine rule. If both sellers experiment, the mean price rises with precision, from the Nash level to that of the consistent conjecture $\lambda = \left(1 - \sqrt{1 - 4\omega^2}\right)/(2\omega)$ (Bresnahan 1981). For small cost dispersion, X earns more than R against R.

The rival's estimates also respond. Its price follows the seller's price at two rates:

WITHIN A PERIOD		IN THE LONG RUN
w_j	$<$	$\dfrac{e}{b + \beta_j}$
the rival discounts each quote for its noise		the rival learns the seller's mean price without error



Profit from mean prices against a rival that acts on the slope b. It peaks at the rival's long-run rate, above the reaction that an experiment measures.

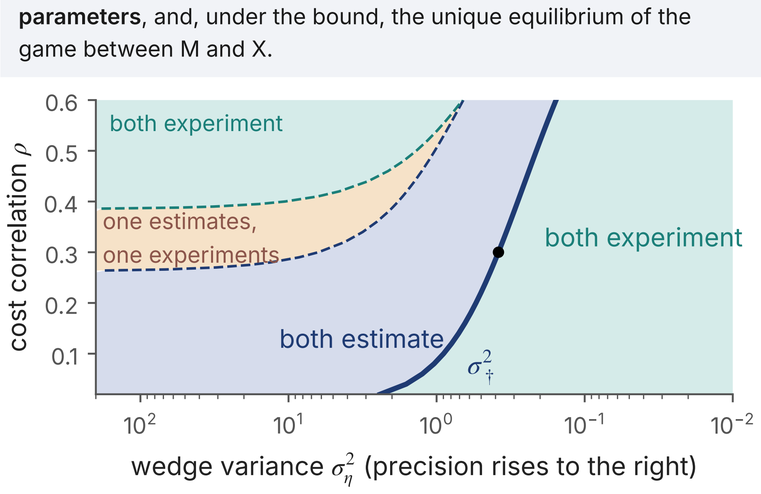
6 Which procedure do sellers adopt?

Theorem 2. With exact quotes, X is strictly dominant under a bound on cost dispersion. With useless quotes and weakly correlated costs, M is strictly dominant.

Theorem 3. Let $w^\dagger$ be the root in $(0, \lambda)$ of

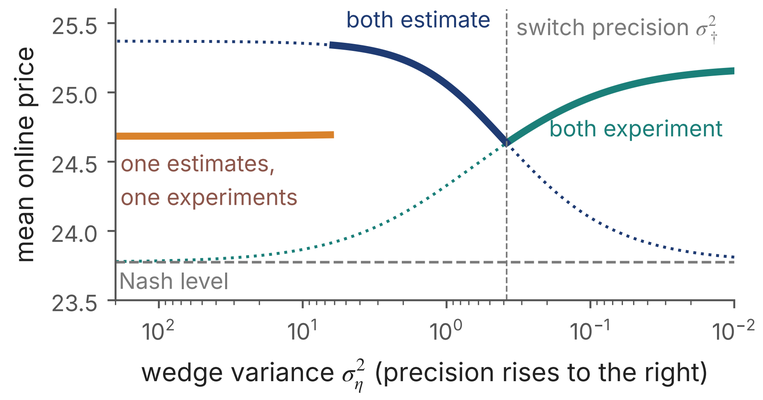
$$w^2\,(1 + \lambda^2 - 2\lambda w) = \rho\,(\lambda - w)(1 - \lambda w),$$

and $1/\sigma_\dagger^2$ the precision at which two experimenting sellers use the quote weight $w^\dagger$. At this **switch precision**, both procedures act on the same conjecture. Just below it, M is a best reply to M **for all parameters**, and, under the bound, the unique equilibrium of the game between M and X.



Equilibrium of the game among M, X, and R, from exact stopping points. Solid curve: the switch precision. Dot: the benchmark. In 336 parameter cells, (X,X) is the unique equilibrium from exact quotes to the switch precision and (M,M) on a band below it. (R,R) is an equilibrium at no interior precision.

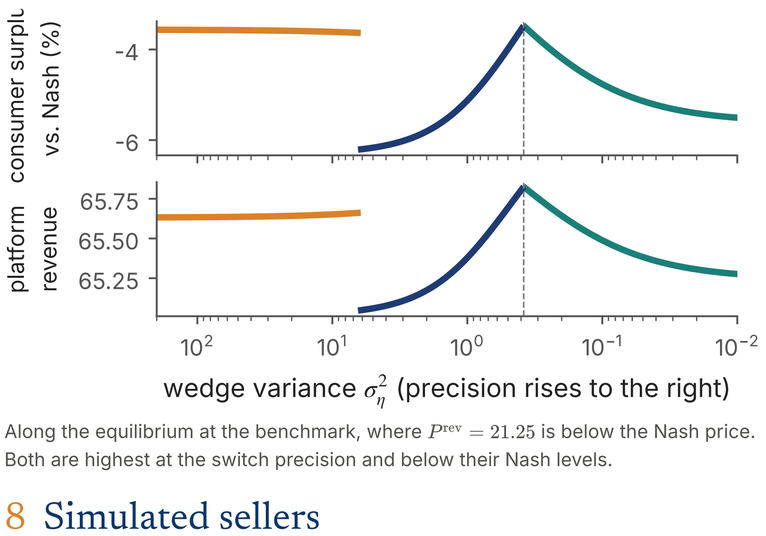
7 Prices and welfare



Mean online price at the benchmark. Thick: equilibrium. Dotted: both estimate, or both experiment, where this is not the equilibrium.

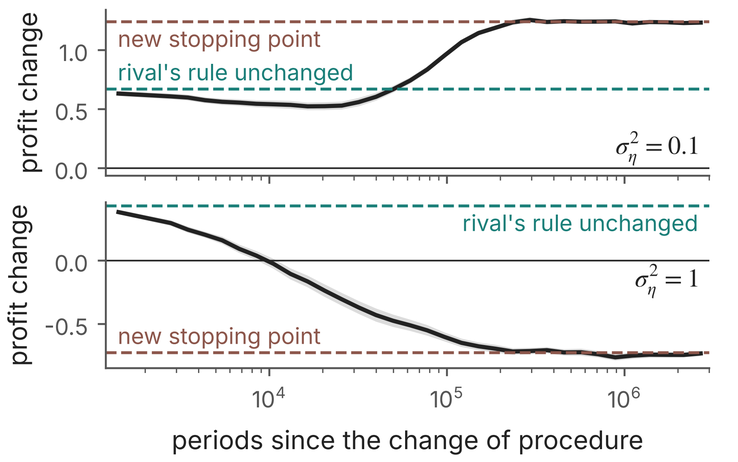
Proposition 2. If $\omega\,(1 + 3\rho^2) \le 2\rho$, then under the bound the equilibrium mean price has a strict local minimum at the switch precision. For small cost dispersion, consumer surplus has a strict local maximum there, and platform revenue a local maximum or minimum according to whether sellers price above or below $P^{\text{rev}} = (\alpha + d\,p^F)/(2(b - e))$.

σ_η^2	equilibrium	mean price	above Nash	consumer surplus	platform revenue
0	(X,X)	25.18	5.9%	−5.6%	65.25
0.38	all four	24.63	3.6%	−3.5%	65.83
1	(M,M)	25.06	5.4%	−5.1%	65.38
5	(M,M)	25.33	6.5%	−6.2%	65.06



Along the equilibrium at the benchmark, where $P^{\text{rev}} = 21.25$ is below the Nash price. Both are highest at the switch precision and below their Nash levels.

8 Simulated sellers



A seller replaces its regression by experiments. Mean change in profit per period over 100 paired runs, with two standard errors. Dashed: predictions.

- Sellers that learn on finite samples reach the predicted stopping points, with slopes within 0.003.
- At a low precision, the change earns 0.30 ± 0.01 more per period at first and 0.74 ± 0.01 less in the long run. The gain lasts about a fifth of the memory of the rival's regression.

9 Implications

- **Sellers.** Whether an unbiased procedure earns more depends on the precision of the rival-price records and on the horizon of the evaluation. A short pilot always favors experimentation.
- **Platforms and regulators.** Under the condition of Proposition 2, a regulator that seeks low prices has an interior target precision.
- **Algorithmic pricing.** Prices above the Nash level arise without communication and from unbiased experiments.

Conclusion

Sellers keep a regression that they know to be biased because the rival's price responds more in the long run than an experiment measures. The price effect of transparency depends on the procedure that sellers adopt.

The three procedures at a glance

procedure	holds fixed	conjecture Λ_i	both use it, exact quotes	both use it, useless quotes	in equilibrium at the benchmark
M estimation	the quote	$t_i\,(w_j + \rho\,g_j/g_i)$	0, the Nash rule	ρ, own slope $b - e\rho$	on the band of estimation, $0.38 \le \sigma_\eta^2 \le 6.29$
X experimentation	nothing	w_j	λ, the consistent conjecture	0, the Nash level	from exact quotes to the switch precision, $\sigma_\eta^2 \le 0.38$
R benchmark, knows b	the rival's price	0	0	0	at no interior precision computed

Benchmark: $\alpha = 30$, $b = 2$, $e = 1$ ($\omega = 0.25$), $d = 0.5$, $\phi = 0.15$, $\bar{c} = 8$, $\sigma_c^2 = 4$, $\rho = 0.3$, $p^F = 25$. Nash mean price 23.77. Beyond the band of estimation, one seller estimates and the other experiments.

Bresnahan TF (1981) *Amer. Econom. Rev.* 71:934–945. Ferreira KJ, Lee BHA, Simchi-Levi D (2016) *M&SOM* 18:69–88. Fisher M, Gallino S, Li J (2018) *Management Sci.* 64:2496–2514.
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Data from a research collaboration with a pharmacy retailer; not publicly available.